Мирзаитов Тимур Вариант №13

1)

$$\int (e^x+4)^3 e^x \,\mathrm{d}x = \int (e^x+4)^3 \,\mathrm{d}e^x = \int (e^x+4)^3 \,\mathrm{d}(e^x+4) = \int (e^x+4)^3 \,\mathrm{d}(e^x+4) = \frac{(e^x+4)^4}{4} + C$$

2)

$$\int \frac{\mathrm{d}x}{1+\sqrt{2x+1}} = \left| \begin{matrix} t^2 = 2x+1 \\ 2t\,\mathrm{d}t = 2\,dx \\ t\,\mathrm{d}t = dx \end{matrix} \right| = \int \frac{t\,\mathrm{d}t}{1+\sqrt{t^2}} = \int \frac{t\,\mathrm{d}t}{1+t} = \int (1 - \frac{1}{1+t})\,\mathrm{d}t = t - \ln|1+t| + C$$

Вернемся к переменной x:

$$t^2 = 2x+1$$
$$t = \sqrt{2x+1}$$

$$t - \ln|1+t| + C = \sqrt{2x+1} - \ln(1+\sqrt{2x+1}) + C$$

3)

$$\int \frac{1}{x\sqrt{\ln^3 x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{\ln^3 x}}\,\mathrm{d}\ln x = \frac{-\ln^{-\frac{1}{2}} x}{\frac{1}{2}} + C = -\frac{2}{\sqrt{\ln x}} + C$$

4)

$$\int (3x+18)2^x\,\mathrm{d}x = \left| \begin{matrix} u = 3x+18, & \mathrm{d}u = 3\mathrm{d}x \\ \mathrm{d}v = 2^x \mathrm{d}x, & v = \dfrac{2^x}{\ln 2} \end{matrix} \right| =$$

$$= (3x+18)\frac{2^x}{\ln 2} - \frac{3}{\ln 2}\int 2^x\,\mathrm{d}x = (3x+18)\frac{2^x}{\ln 2} - \frac{3\cdot 2^x}{\ln^2 2} =$$

5)

$$\int \frac{-13x-32}{(x+2)(x-1)(x+4)}\,\mathrm{d}x =$$

$$= \left| \begin{matrix} \dfrac{A}{(x+2)} + \dfrac{B}{(x-1)} + \dfrac{C}{(x+4)} = \dfrac{-13x-32}{(x+2)(x-1)(x+4)} \\ A(x-1)(x+4) + B(x+2)(x+4) + C(x+2)(x-1) = -13x-32 \\ A(x^2+3x-4) + B(x^2+6x+8) + C(x^2+x-2) = -13x-32 \\ Ax^2+3Ax-4A+Bx^2+6Bx+8B+Cx^2+Cx-2C = -13x-32 \\ \begin{matrix} x^2 \\ x^1 \\ x^0 \end{matrix} \begin{matrix} A+B+C & = 0 \\ 3A+6B+C & = -13 \\ -4A+8B-2C & = -32 \end{matrix} \iff \begin{cases} 15B = -45 \\ A+C = -B \\ 3A+C = -6B-13 \end{cases} \iff \begin{cases} B = -3 \\ A+C = 3 \\ 3A+C = 5 \end{cases} \iff \begin{cases} B = -3 \\ A = 1 \\ C = 2 \end{cases} \end{matrix} \right|$$

$$= \int (\frac{1}{(x+2)} - \frac{3}{(x-1)} + \frac{2}{(x+4)})\,\mathrm{d}x = \ln|x+2| - 3\ln|x-1| + 2\ln|x+4| + C$$

6)

$$\int \frac{(x+1)^3}{x^2}\,\mathrm{d}x = \int \frac{x^3+3x^2+3x+1}{x^2}\,\mathrm{d}x = \int (x+3+\frac{3}{x}+\frac{1}{x^2})\,\mathrm{d}x =$$
$$= \frac{x^2}{2}+3x+3\ln|x|-\frac{1}{x}+C$$

7)

$$\int \frac{1}{x\sqrt{1-\ln^2 x}}\,\mathrm{d}x = \int \frac{1}{\sqrt{1-\ln^2 x}}\,\mathrm{d}\ln x = \arcsin(\ln x)+C$$

8)

$$\int \frac{1}{(1+\sqrt{x})^2}\,\mathrm{d}x = \left|\begin{matrix} t^2=x \\ 2t\,\mathrm{d}t=\mathrm{d}x \end{matrix}\right| = \int \frac{2t}{(1+t)^2}\,\mathrm{d}t = 2\int \frac{t}{(1+t)^2}\,\mathrm{d}t = 2\int \frac{t+1-1}{(t+1)^2}\,\mathrm{d}t =$$
$$= 2\int \frac{1}{t+1}-\frac{1}{(t+1)^2}\,\mathrm{d}t = 2\ln|t+1|+\frac{2}{t+1}$$

Вернемся к переменной x:

$$t^2=x$$
$$t=\sqrt{x}$$

$$2\ln|t+1|+\frac{2}{t+1} = 2\ln|\sqrt{x}+1|+\frac{2}{\sqrt{x}+1}$$

9)

$$\int \frac{\mathrm{d}x}{5+5\cos^2 x+sin^2 x} =$$
$$= \left|\begin{matrix} t=\tan x \\ \mathrm{d}x=\dfrac{\mathrm{d}t}{1+t^2} \\ \cos x=\dfrac{1}{\sqrt{1+t^2}} \\ \sin x=\dfrac{t}{\sqrt{1+t^2}} \end{matrix}\right| = \int \frac{\frac{\mathrm{d}t}{1+t^2}}{5+5(\frac{1}{\sqrt{1+t^2}})^2+(\frac{t}{\sqrt{1+t^2}})^2} = \int \frac{\frac{\mathrm{d}t}{1+t^2}}{5+\frac{5}{1+t^2}+\frac{t^2}{1+t^2}} =$$
$$= \int \frac{\frac{\mathrm{d}t}{1+t^2}}{\frac{5+5t^2+5+t^2}{1+t^2}} = \int \frac{\mathrm{d}t}{6t^2+10} = \frac{1}{6}\int \frac{\mathrm{d}t}{t^2+\frac{5}{3}} = \frac{1}{6}\frac{1}{\sqrt{\frac{5}{3}}}\arctan\frac{t}{\sqrt{\frac{5}{3}}}+C =$$
$$= \frac{1}{6}\sqrt{\frac{3}{5}}\arctan t\sqrt{\frac{3}{5}}+C = \frac{1}{6}\sqrt{\frac{3}{5}}\arctan(\tan x\sqrt{\frac{3}{5}})+C$$

10)

$$\int \frac{x^2\,\mathrm{d}x}{\sqrt{25-x^2}} =$$

$$= \left| \begin{matrix} x = 5\cos t \\ \mathrm{d}x = -5\sin t\,\mathrm{d}t \end{matrix} \right| = -\int \frac{25\cos^2 t \cdot 5\sin t\,\mathrm{d}t}{\sqrt{25-25\cos^2 t}} = -\int \frac{25\cos^2 t \cdot 5\sin t\,\mathrm{d}t}{5\sqrt{1-\cos^2 t}} =$$

$$= -\int \frac{25\cos^2 t \cdot 5\sin t\,\mathrm{d}t}{5\sqrt{\sin^2 t}} = -\int \frac{25\cos^2 t \cdot 5\sin t\,\mathrm{d}t}{5\sin t} = -25\int \cos^2 t\,\mathrm{d}t$$

$$= -25\int \frac{1+\cos 2t}{2}\,\mathrm{d}t = -\frac{25}{2}\int (1+\cos 2t)\,\mathrm{d}t = -\frac{25}{2}t - \frac{25}{4}\sin 2t + C =$$

$$= -\frac{25}{2}\arccos\left(\frac{x}{5}\right) - \frac{25}{4}\sin 2(\arccos\frac{x}{5}) + C$$